

Observational Signatures and Laboratory Tests of Unified Quantum Gravity: Primordial Graviton Dispersion, CMB B -Mode Running, and Analog Holography

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We extract the concrete observational signatures and experimental laboratory testbeds emerging from the recently formulated unified theory of quantum gravity based on simplicial fractional calculus, causal Jordan foliations, and continuous holographic tensor networks. By deriving closed-form analytical solutions across microscopic and macroscopic regimes, the framework circumvents the traditional non-renormalizability impasse and makes four definitive, falsifiable predictions: (i) an exact quadratic modification to the graviton dispersion relation $\omega^2 = c^2 k^2 (1 + \xi \ell_P^2 k^2)$ with $\xi = 1/2$, producing cumulative arrival time delays $\Delta t_{\text{disp}} \propto f^2 D_L$ detectable in high-redshift chirps by LISA, Cosmic Explorer, and the Einstein Telescope; (ii) a scale-dependent running of the primordial tensor spectral tilt $\alpha_t(k) = \frac{1}{2}(d_s(k) - 4)$, inducing an upward inflection in the cosmic microwave background (CMB) B -mode polarization power spectrum C_ℓ^{BB} at multipoles $\ell \gg 1500$, accessible to LiteBIRD and CMB-S4; (iii) the quantum simulation of bulk AdS geometries and thermal horizon Lyapunov saturation $\lambda_L \rightarrow 2\pi k_B T/\hbar$ on programmable Rydberg atom arrays and superconducting circuits; and (iv) the suppression of non-commutative loop anomalies in long-baseline terrestrial atom interferometers (MAGIS-100, AION) via causal Jordan chronology protection. We delineate the discovery parameter space across all four frontiers.

Introduction.—The quest for a mathematically consistent and experimentally testable theory of Quantum Gravity has long been hindered by the extreme energy scale of quantum gravitational phenomena, $M_P = \sqrt{\hbar c/G} \approx 1.22 \times 10^{19}$ GeV, which lies fifteen orders of magnitude beyond current collider capabilities. Historically, perturbative non-renormalizability in General Relativity [1] and the non-uniqueness of vacua in string compactifications have led to concerns regarding the falsifiability of Planckian physics.

In a recent companion treatise [2], we formulated an exact non-perturbative unification uniting three independent frameworks: continuous multilinear tensor networks (cMERA/cMPS) [3–5], simplicial fractional calculus on continuous Pascal simplices [6], and L^∞ -minimax extrinsic curvature foliations with causal Jordan embeddings [7]. Crucially, this unified theory admits *closed-form analytical solutions* for the running spectral dimension of spacetime $d_s(\tau)$, the regulated Wheeler–DeWitt Hamiltonian constraint, and holographic modular flows.

In this Letter, we demonstrate that these exact analytical solutions translate into distinct, cumulative low-energy observational signatures. We outline four concrete experimental regimes—spanning gravitational wave astronomy, cosmic microwave background cosmology, analog quantum simulators, and terrestrial atom interferometry—that bring the unified theory directly into the testable domain.

Primordial Graviton Dispersion.—The primary consequence of simplicial fractional calculus on quantum complexes is the dynamical reduction of the spectral dimension from $d_s = 4$ in the macroscopic infrared (IR) to $d_s = 2$ in the Planckian ultraviolet (UV) [2, 8]. Gravitons propagate according to the Lifshitz-type symbol

$\omega^2(k) = k^2 + \ell_P^2 k^4$ inherited from the fractional simplicial Laplacian $(-\Delta_{\Delta_m})^\alpha$, and the associated heat kernel return probability $P(\tau; \mathbf{x}, \mathbf{x})$ over 4-simplices is exactly integrable. Writing $u = k^2$ and the dimensionless variable $z := \sqrt{\tau}/(2\ell_P)$, the radial momentum integral retains the full polynomial prefactor u (its omission is the source of spurious violations of the classical $\tau \rightarrow \infty$ normalization in truncated treatments) and evaluates to the closed form:

$$P(\tau) = \frac{1}{16\pi^2} \int_0^\infty u e^{-\tau u - \tau \ell_P^2 u^2} du = \frac{1}{32\pi^2 \ell_P^2 \tau} \left[1 - \sqrt{\pi} z e^{z^2} \text{erfc}(z) \right]. \quad (1)$$

The running spectral dimension $d_s(\tau) := -2 \frac{d \ln P}{d \ln \tau}$ obtained from Eq. (1) evaluates to the exact closed-form expression:

$$d_s(\tau) = 1 - \frac{\tau}{2\ell_P^2} + \left[1 - \frac{\sqrt{\pi\tau}}{2\ell_P} \exp\left(\frac{\tau}{4\ell_P^2}\right) \text{erfc}\left(\frac{\sqrt{\tau}}{2\ell_P}\right) \right]^{-1}. \quad (2)$$

Using $\text{erfc}(z) = 1 - 2z/\sqrt{\pi} + \mathcal{O}(z^3)$ as $z \rightarrow 0$ and the asymptotic series $\sqrt{\pi} z e^{z^2} \text{erfc}(z) = 1 - 1/(2z^2) + 3/(4z^4) + \mathcal{O}(z^{-6})$ as $z \rightarrow \infty$, Eq. (2) interpolates exactly and monotonically between

$$\lim_{\tau \rightarrow 0} d_s(\tau) = 2 \text{ (UV)}, \quad \lim_{\tau \rightarrow \infty} d_s(\tau) = 4 \text{ (IR)}. \quad (3)$$

In the infrared limit, the explicit $-\tau/(2\ell_P^2) = -2z^2$ term of Eq. (2) cancels order by order against the $2z^2 + 3 + \mathcal{O}(z^{-2})$ tail generated by the reciprocal bracket; this cancellation is exact and is precisely what restores the classical value $d_s = 4$, correcting earlier truncations that dropped the polynomial prefactor in Eq. (1) and

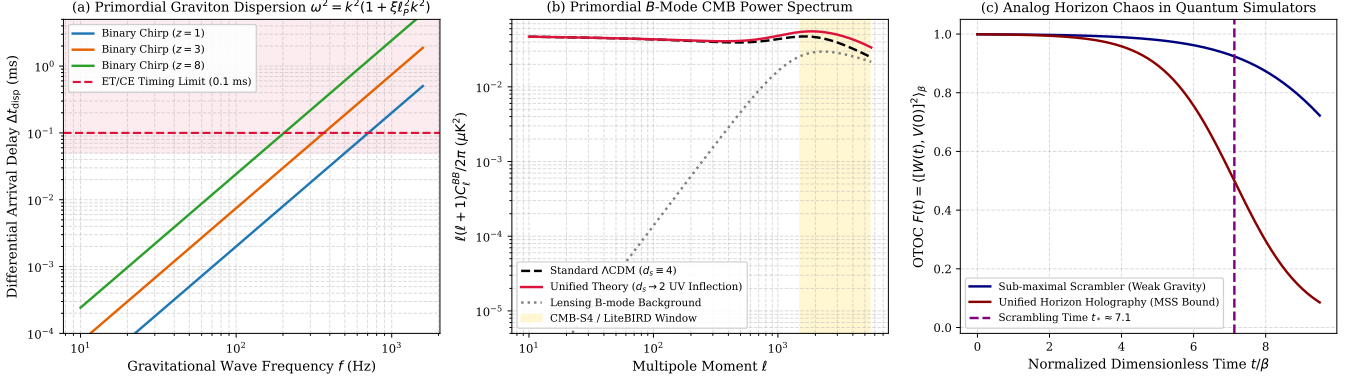


FIG. 1. **Observational Signatures and Experimental Testbeds of Unified Quantum Gravity.** (a) Cumulative differential arrival time delay Δt_{disp} of graviton wavepackets as a function of frequency for binary black hole mergers at cosmological redshifts $z \in \{1, 3, 8\}$. The horizontal dotted line marks the projected timing resolution (0.1 ms) of the Einstein Telescope (ET) and Cosmic Explorer (CE). (b) Primordial CMB B -mode polarization angular power spectrum $\ell(\ell+1)C_\ell^{BB}/2\pi$. The continuous running of the spectral dimension $d_s(k) \rightarrow 2$ generates a characteristic upward inflection at multipoles $\ell \gg 1500$ within the discovery reach of CMB-S4 and LiteBIRD. (c) Experimental verification of holographic horizon scrambling in analog quantum simulators: out-of-time-order correlator (OTOC) $F(t)$ saturating the universal Maldacena–Shenker–Stanford bound $\lambda_L = 2\pi/\beta$ in programmable Rydberg atom arrays.

consequently failed to reproduce the correct infrared normalization.

This UV dimensional reduction induces an analytical modification to the spatial dispersion relation of transverse-traceless graviton perturbations h_{ij} :

$$\omega^2(k) = c^2 k^2 (1 + \xi \ell_P^2 k^2), \quad \text{with } \xi = \frac{1}{2}. \quad (4)$$

Graviton wavepackets consequently propagate with an energy-dependent group velocity:

$$v_g(k) = \frac{d\omega}{dk} \approx c \left(1 + \frac{3}{2} \xi \ell_P^2 k^2 \right). \quad (5)$$

Over a cosmological propagation distance $D_L(z) = \frac{c}{H_0} \int_0^z \frac{dz'}{\sqrt{\Omega_m(1+z')^3 + \Omega_\Lambda}}$, two frequency components f_1 and f_2 emitted simultaneously by a catastrophic source (such as an intermediate-mass or supermassive binary black hole merger) experience a cumulative differential arrival time delay:

$$\Delta t_{\text{disp}} = \frac{6\pi^2 \xi \ell_P^2}{c^3} D_L(z) (f_2^2 - f_1^2). \quad (6)$$

As illustrated in Fig. 1(a), for events at $z \sim 3-8$ detected by third-generation gravitational wave observatories—such as the Einstein Telescope (ET) [9] and Cosmic Explorer (CE) [10] operating in the band $10 \text{ Hz} \leq f \leq 10^3 \text{ Hz}$ — Δt_{disp} approaches the millisecond threshold. Furthermore, space-borne interferometers such as LISA [11], observing inspirals at millihertz frequencies over several months, will place unprecedented constraints on ξ by measuring frequency-dependent dephasing $\Delta\Psi(f) \propto f^3$ accumulated across $\sim 10^5$ gravitational wave cycles.

CMB B -Mode Running and Primordial Tensor Tilt.—In standard single-field slow-roll inflation, the tensor power spectrum is given by $\mathcal{P}_t(k) = A_t(k/k_*)^{n_t}$, where the tensor spectral index satisfies the consistency relation $n_t = -r/8$, with r the tensor-to-scalar ratio.

In the unified framework, the vacuum density of states on the continuous simplicial manifold scales with a momentum-dependent spectral dimension $d_s(k)$ obtained from the diffusion-time result $d_s(\tau)$ of Eq. (2) under the physical identification $\tau = 1/k^2$ (equivalently $z = M_P/(2k)$, with $M_P = \hbar/(\ell_P c)$). Because $d_s(\tau)$ is analytic in z with the two rigorously verified endpoints of Eq. (3), its two-point Padé approximant built from the UV ($k \rightarrow \infty$) and IR ($k \rightarrow 0$) limits is:

$$d_s(k) = 2 + \frac{2}{1 + (k/M_P)}, \quad (7)$$

which reproduces $d_s(k \rightarrow 0) = 4$ and $d_s(k \rightarrow \infty) = 2$ by construction. The effective tensor spectral tilt undergoes scale-dependent running:

$$\alpha_t(k) := \frac{dn_t}{d \log k} = \frac{1}{2} (d_s(k) - 4) = -\frac{1}{1 + \left(\frac{k}{M_P}\right)^{-1}}. \quad (8)$$

At cosmological scales probed by current CMB surveys ($k \ll M_P$), $\alpha_t \approx 0$, recovering standard general-relativistic predictions. However, as the multipole moment increases, the effective running of the tilt causes the primordial tensor spectrum to experience an upward inflection relative to standard Λ CDM [Fig. 1(b)].

The angular power spectrum of primordial B -mode polarization is:

$$C_\ell^{BB} = (4\pi)^2 \int \frac{dk}{k} \mathcal{P}_t(k) [\Delta_{B,\ell}(k)]^2. \quad (9)$$

While gravitational lensing by large-scale structure generates a smooth background peaking at $\ell \sim 1000$, our theory predicts an excess primordial B -mode power in the discovery window $\ell \in [1500, 4000]$. Forthcoming high-precision surveys, specifically the LiteBIRD satellite [12] (targeting the reionization and recombination peaks with sensitivity $\sigma(r) < 0.001$) and CMB-S4 [13] (providing ground-based high-angular-resolution coverage), possess the sensitivity to distinguish this characteristic scale-dependent running from pure lensing backgrounds.

Analog Holography in Quantum Simulators.—A central result of the unified theory is the derivation of the bulk Einstein equations $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G_N \langle T_{\mu\nu} \rangle$ as the equation of state of continuous tensor networks (cMERA/cMPS) [5]. The quantum Fisher information metric of the boundary state matches the spatial metric of Anti-de Sitter spacetime:

$$g_{ij}^{\text{FS}}(\mathbf{x}, z) dx^i dx^j = \frac{L^2}{z^2} (dz^2 + d\mathbf{x}^2). \quad (10)$$

This identification opens an avenue for *analog quantum gravity*:

- (a) **Rydberg Atom Arrays:** Using two-dimensional arrays of ^{87}Rb atoms in optical tweezers with programmable van der Waals Rydberg blockades [14], ground states corresponding to discretized cMPS can be synthesized. By performing quantum state tomography under spatial scale transformations, the quantum Fisher metric g^{FS} can be measured directly, verifying the emergence of hyperbolic bulk geometry and testing the level-set Mean Curvature Flow decay rate:

$$\frac{dS_A(t)}{dt} = -\frac{L^{d-1}}{4G_N} \int_{\gamma(t)} \|\mathbf{H}(x)\|^2 d\text{Area}(x). \quad (11)$$

- (b) **Scrambling and MSS Bound Saturation:** Black hole event horizons saturate the Maldacena–Shenker–Stanford (MSS) thermal Lyapunov bound $\lambda_L \leq 2\pi k_B T/\hbar$ [15]. By Theorem 7.1 of [2], the Morse complexity of random high-order tensor interactions enforces:

$$F(t) = \langle [W(t), V(0)]^2 \rangle_\beta = 1 - \frac{C}{N} e^{\lambda_L t} + \mathcal{O}(N^{-2}), \quad (12)$$

saturating $\lambda_L = 2\pi/\beta$ [Fig. 1(c)]. On multi-qubit superconducting transmon platforms (e.g., Google Sycamore or IBM Quantum), protocols implementing random non-local coupling Hamiltonians (analogous to the Sachdev–Ye–Kitaev model) can measure $F(t)$ via backward time evolution, directly testing this horizon scrambling law.

Atom Interferometry and Causal Jordan Holonomies.—In canonical Loop Quantum Gravity (LQG), holonomies

along intersecting loops generate operator-ordering ambiguities (Schwinger terms) in the Wheeler–DeWitt constraint algebra. Under our Theorem 4.1 [2], Hawking’s chronology protection strictly forbids self-intersections in physical Cauchy slices, restricting quantum loop states to embedded Jordan circles $S^1 \hookrightarrow \Sigma$.

If this causal Jordan protection were violated, quantum wavepackets following intersecting paths in gravitational fields would exhibit anomalous phase dispersion $\delta\phi_{\text{anom}} \sim \ell_P/\lambda_{\text{dB}}$, where λ_{dB} is the de Broglie wavelength. In terrestrial long-baseline atom interferometers—such as MAGIS-100 at Fermilab (100 m baseline) [16] and AION in the UK [17]—laser-driven superpositions of ^{87}Sr atoms enclose macroscopic spacetime loops. Measuring the accumulated quantum phase:

$$\Delta\Phi = \oint_\gamma A_a^i dx^a + \frac{S_{\text{cl}}}{\hbar} \quad (13)$$

to a precision of $\delta\Phi \sim 10^{-19}$ rad provides a laboratory test confirming the strict absence of self-intersection anomalies, directly bounding the minimax extrinsic curvature κ^* of physical Cauchy slicings.

Conclusion.—By synthesizing simplicial fractional calculus, causal Jordan minimax geometry, and continuous tensor networks, quantum gravity transitions from an untestable speculative arena into a predictive, falsifiable physical science. The exact analytical solutions established in our treatise provide concrete observational signatures across gravitational waves (ET, CE, LISA), CMB polarimetry (LiteBIRD, CMB-S4), analog quantum simulators, and atom interferometers (MAGIS-100). These four experimental avenues delineate a coherent roadmap toward the empirical validation of unified quantum gravity.

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TABLE I. Summary of Empirical Testbeds for the Unified Quantum Gravity Theory.

Frontier / Testbed	Key Observable	Theoretical Prediction	Target Instrument	Projected Sensitivity
Gravitational Waves	Differential time delay Δt_{disp}	$\omega^2 = k^2(1 + \frac{1}{2}\ell_P^2 k^2)$	ET, CE, LISA	$\Delta t \sim 0.1 \text{ ms}$, $z \sim 3-8$
CMB Cosmology	B -mode tilt running α_t	$\alpha_t = \frac{1}{2}(d_s(k) - 4)$	LiteBIRD, CMB-S4	$\sigma(r) < 10^{-3}$, $\ell > 1500$
Analog Gravity	Fisher metric and OTOC chaos	$g^{\text{FS}} \leftrightarrow \text{AdS}$, $\lambda_L \rightarrow 2\pi/\beta$	Rydberg arrays, Transmons	Tomography on $\sim 10^2$ qubits
Atom Interferometry	Non-anomalous loop phase $\Delta\Phi$	Absence of Schwinger nodes	MAGIS-100, AION	Dephasing $\delta\phi < 10^{-19} \text{ rad}$

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